Ege Hurturk

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Education

Imperial College London **Expected Graduation: June 2027**
BEng Computing - Expected First Class (First year grade: %78.42) *London, UK*

- Second Year Courses: Machine Learning, Operating Systems, Networks and Communications, Algorithm Design and Analysis, Models of Computation, Computer Practical 2, Software Engineering Design, Probability and Statistics.

Work Experience

AwesomeMotive **Jun 2023 – Nov 2023**
Research and Development Intern *West Palm Beach, FL (Remote)*

- Developed advanced transformer models for context-aware spam detection of blog content, improving detection accuracy.
- Architected and contributed to SeedProdAI, a system that generates and deploys content-ready WordPress sites from user descriptions using AI-driven templating.
- Reduced site launch times by over 50% (from 4 minutes to under 2) through a custom Go-based provisioning tool that parallelized server deployment and AI content generation via concurrent execution.

Projects

WACC **Scala | AArch64 Assembly | Compiler Design**

- Built a multi-phase compiler featuring lexer, parser, renamer, and type checker for producing a type-parameterised AST, and flattener, register allocator, and code generator, for emitting AArch64 assembly via a two-tier IR design.
- Extended the language with a local static analyser that catches errors at compile-time, removing redundant runtime checks for provably safe operations, and optimised generated IR with convergent peephole and tail-recursion optimisations.
- Employed unit, integration, and randomised property-based testing with GitLab CI for continuous integration.

PintOS **C | x86 Assembly | Concurrent Programming**

- Implemented core OS subsystems including preemptive scheduling (priority donation, BSD scheduler) and a full system call interface for process control, file I/O, and safe user memory access
- Designed and integrated a virtual memory system with supplemental page tables, frame allocation, eviction (swapping), lazy loading, and memory-mapped files to improve memory utilisation and I/O performance
- Engineered concurrent kernel components with careful synchronisation, debugging race conditions using GDB/QEMU, and passing an extensive test suite covering process management and VM behaviour

ARMv8 Assembler Emulator | *GitHub* **C | ARMv8 Assembly | CMake**

- Built a two-pass assembler for ARMv8 in C, with lexical parsing and symbol resolution, supporting ASM instructions.
- Developed an emulator with full instruction decoding, register simulation, and a memory model.
- Implemented a minimal neural network library from scratch in C, training a multi-layer perceptron for digit classification.

TSNet | *GitHub* **WaveNet | PyTorch | DSP | Python**

- Trained a WaveNet architecture for modeling the Tube Screamer pedal, achieving an error2signal ratio loss of 0.079.
- Collected and labeled dataset of 640 seconds of audio input.
- Wrote a paper on investigating the correlation between audio input length and the performance of the WaveNet model.

Skills, Activities & Interests

Languages: Python, C, C++, Scala, Java, Haskell, Kotlin, SQL, Swift

Developer Tools: AWS, PostgreSQL, PlanetScale, Redis, Celery, Kafka, ElasticSearch, GCP, Postman, SupaBase, Git, Azure, Docker

Libraries/Frameworks: PyTorch, TensorFlow, FastAPI, SwiftUI, Spring Boot, ReactJS, NextJS, Django

Interests: I am the lead guitarist of Red Rosary, and we're working on releasing our first album!